

Capstone Project Proposal

Organization Name:

Virtual Gold Inc (AI Assistant)

Organization Address (if same as Parent Org Address, please re-enter):

3399 Shore PKWY Brooklyn, NY 11235

Briefly Describe Your Organization:

Founded by Inderpal Bhandari in 1997, Virtual Gold is a privately-owned firm specializing in Business, Data, and AI Consulting services for enterprises, alongside offering bespoke mentorship programs tailored to the specific needs of start-up companies.

Organization Website:

<https://www.virtualgold.co/>

Capstone Project Overview**Please be as specific as you can in communicating your proposal description.:**

This project will design and evaluate a secure AI assistant that leverages open-source large language models running locally while selectively utilizing more capable cloud-based models when additional confidence or reasoning capability is required. The platform will prioritize privacy, security, performance, and cost efficiency by keeping sensitive data local whenever possible. Students will investigate architectures, routing mechanisms, security controls, and performance optimization techniques that enable organizations to deploy trustworthy AI assistants in enterprise environments.

Describe the Capstone Problem:

Organizations increasingly want to deploy AI assistants but face concerns regarding data privacy, security, cost, model reliability, and vendor dependence. While open-source models offer greater control and privacy, they may not always provide sufficient performance for complex tasks. Conversely, cloud-based models often deliver superior capabilities but introduce security, compliance, and cost considerations. Students will investigate how to build a hybrid AI architecture that uses local models as the primary intelligence layer, measures confidence in generated responses, and selectively escalates requests to more capable cloud-based models when appropriate. The project will also evaluate security risks associated with open-source and international AI models and identify safeguards for protecting enterprise data and operations.

Objectives of the Project:

The primary objective is to design and evaluate a secure, high-performance hybrid AI assistant architecture that balances privacy, performance, cost, and model capability.

Students may investigate:

- Open-source LLM deployment and optimization
- Confidence scoring and model-routing strategies
- Hybrid local/cloud AI architectures
- Security and privacy controls
- AI governance and risk management
- Benchmarking and performance evaluation
- Safeguards for using third-party and international AI models

Deliverables may include system architecture designs, prototype implementations, benchmarking frameworks, security assessments, performance analyses, and

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recommendations for enterprise deployment. Results will help Virtual Gold evaluate practical approaches for secure and cost-effective enterprise AI adoption.

What is the business value of the project to your organization?:

This project aligns with Virtual Gold's focus on helping organizations adopt AI responsibly while maximizing business value. The project will provide insights into secure AI deployment strategies that reduce reliance on external providers, improve data privacy, and optimize operating costs. The findings will help identify best practices for hybrid AI architectures, confidence-based model routing, and enterprise AI governance. The results may support future consulting services, AI transformation initiatives, executive education programs, and thought leadership related to secure and trustworthy AI adoption.

Will your organization provide data to the student team? :

Yes

What type of data will you provide for the students? We request that you have all data ready to share with students at the beginning of the project. Check all that apply::

Public Data

Confirm that the data you share will contain NO Personally Identifiable Information (PII):

Confirmed

Public Data: Describe at a high level, descriptions of the data you indicated above. Write N/A if you are not providing this type of data. :

Examples of enterprise and small business data that are publicly available which can be used to create synthetic data sets as needed. We will only be providing publicly available data.

Skills & Technologies Required:

The students will be working on their laptops/desktops using publicly available or synthetic data. They will be working on a sandboxed environment. Interfacing with corporate systems is not anticipated at this time.

Primary Capstone Lead Contact Name:

Urte Jesina

Capstone Lead Contact Job Title:

Executive Assistant to Inderpal Bhandari

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